Cargo

V1 module guide: what the system is made of and what each part does

Kigali, Rwanda | October 2026 | Prepared by Christian, founder

1. Overview

Cargo V1 is built as a set of modules. Each module owns one clear job (for example lots, grading or payments), has its own folder in the backend, and exposes its own part of the web and mobile applications. Building by module keeps the code organised, lets work be tested one piece at a time, and makes it possible to improve or replace one part without breaking the others.

This guide lists every module needed for V1, grouped by purpose. For each module it explains what it is for, what it does, who uses it, what data it keeps, what it depends on, and which requirements in the Software Requirements Specification (SRS) it satisfies. Items marked (Should) or (Could) are not required for the first release.

There are 16 business modules and 6 supporting platform services. The business modules fall into four groups:

Group	Modules	What the group does
Foundation	Authentication, Users and roles, Organisations and verification, Catalogue and settings, Audit log	Who people are, what they may do, and the rules and records everything else relies on
Marketplace	Lots, Grading and quality, Requests, Offers and matching, Orders	The trading flow from listed stock to an agreed order
Money and fulfilment	Payments and settlement, Deliveries and logistics, Invoicing and records, Disputes	Getting paid, moving goods, producing paperwork and resolving problems
Communication and insight	Notifications, Reports and dashboards	Keeping people informed and showing how the business is doing

2. Module map

The table shows who sees each module on screen. Every module also has a backend part. Yes means the role has real screens for it, Limited means they only view or choose from it, and a dash means they do not use it.

#	Module	Group	Buyer (web)	Admin (web)	Supplier (mobile)	Field agent (mobile)
1	Authentication	Foundation	Yes	Yes	Yes	Yes
2	Users and roles	Foundation	Limited	Yes	Limited	Limited
3	Organisations and verification	Foundation	Yes	Yes	Yes	Yes
4	Catalogue and settings	Foundation	Limited	Yes	Limited	Limited
5	Audit log	Foundation	-	Yes	-	-
6	Lots	Marketplace	Yes	Yes	Yes	Limited
7	Grading and quality	Marketplace	Limited	Yes	Limited	Yes
8	Requests	Marketplace	Yes	Yes	-	-
9	Offers and matching	Marketplace	Yes	Yes	-	-
10	Orders	Marketplace	Yes	Yes	Yes	-
11	Payments and settlement	Money and fulfilment	Yes	Yes	Yes	-
12	Deliveries and logistics	Money and fulfilment	Yes	Yes	Limited	-
13	Invoicing and records	Money and fulfilment	Yes	Yes	Yes	-
14	Disputes	Money and fulfilment	Yes	Yes	Yes	-
15	Notifications	Communication and insight	Yes	Yes	Yes	Yes
16	Reports and dashboards	Communication and insight	Limited	Yes	Limited	-

3. Business modules in detail

Foundation

1. Authentication

Backend folder: **auth** | Priority: **Must**

Proves who a user is and keeps accounts safe. Every other module relies on it to know who is acting and what they may do.

Main capabilities	• Registration request that waits for administrator approval • Login with phone number or email and password • One-time password (OTP) at first login and on a new device • Password reset by OTP • Temporary lock after repeated failed logins • Secure sessions that expire after inactivity; sign out of all devices (Should) • Secure token storage on the phone
Used by	Web app, mobile app and backend (all roles)
Key data	Password hashes, OTP codes, sessions and tokens, device records, login attempts
Depends on	Users, Notifications (to send OTPs), Integrations (SMS and email)
SRS requirements	FR-AUTH-01 to FR-AUTH-08, NFR-SEC-01, NFR-SEC-03, NFR-SEC-04, NFR-SEC-09

2. Users and roles

Backend folder: **users** | Priority: **Must**

Manages user accounts and decides what each kind of user is allowed to see and do.

Main capabilities	• Four roles: supplier, field agent, buyer, administrator • Permissions enforced on the server for every request • Approve, reject, suspend and reactivate accounts • Administrator sub-roles such as operations, finance and support (Should) • Language preference (Kinyarwanda or English) and notification preferences
Used by	Administrators manage users; every user edits their own profile and language
Key data	User, role, permission, account status, language, last login
Depends on	Authentication, Organisations
SRS requirements	FR-AUTH-02, FR-AUTH-07, FR-ADM-01, FR-ADM-06, NFR-SEC-02

3. Organisations and verification

Backend folder: **organisations** | Priority: **Must**

Holds the profiles of cooperatives (suppliers) and buying institutions, and makes sure only verified suppliers can sell.

Main capabilities	• Supplier profile: cooperative name, registration number, district and sector, contact person, crops, approximate capacity • Buyer profile: organisation type, registration and tax details, delivery locations, contacts • Verification status: pending, verified, suspended • Field agent visit records with notes, photos and location • Administrator approval after reviewing details and documents • Several named users for one organisation (Should) • Trade history and rating summary (Should)
Used by	Suppliers, buyers and field agents (own profile and visits); administrators (approval)
Key data	Organisation, contacts, documents, visits, verification status, trade summary
Depends on	Users, File and media storage
SRS requirements	FR-ORG-01 to FR-ORG-07

4. Catalogue and settings

Backend folder: **crops, settings** | Priority: **Must**

Holds the reference data and rules that the marketplace runs on, so that administrators can change them without a developer.

Main capabilities	• Crop catalogue, varieties and units (kilograms, tons) • Districts and sectors • Grading templates and grade rules for each crop • Fee rates: commission and inspection fee • Delivery tolerances, expiry periods and order terms text • Settings changes are recorded in the audit log
Used by	Administrators edit; other modules and apps read
Key data	Crop, unit, district, grading template, grade rule, fee setting, tolerance, expiry period
Depends on	Audit log
SRS requirements	FR-ADM-02, FR-QA-01, FR-ORD-06, FR-LOT-07, FR-REQ-09, FR-DEL-04

5. Audit log

Backend folder: **audit** | Priority: **Must**

Keeps a permanent record of important actions so that any decision or change can be explained later. It is essential for trust, disputes and compliance.

Main capabilities	• Records who did what, to which record, when, with the values before and after • Cannot be edited or deleted • Searchable and filterable by administrators • Called automatically by the other modules
Used by	Administrators (viewing); all modules (writing)
Key data	User, action, object, before and after values, time
Depends on	None (other modules depend on it)
SRS requirements	FR-ADM-05, BR-07

Marketplace

6. Lots

Backend folder: **lots** | Priority: **Must**

Lets suppliers list harvest-ready produce for sale, and controls when a listing becomes visible to buyers.

Main capabilities	• Create a lot: crop, variety, quantity, price expectation, harvest and availability dates, pickup location, photos • Harvest-ready stock only; crops not yet harvested cannot be listed • Statuses: Draft, Submitted, Verified, Reserved, Sold, Rejected, Withdrawn, Expired • Edit or withdraw until verified; later changes need administrator approval • Partial reservation and sale of a lot with remaining quantity tracked • Unique lot code used on every related record • Automatic expiry with a warning first (Should); offline drafts on the phone (Should)
Used by	Suppliers (create), field agents (assigned lots), administrators (review), buyers (verified lots only)
Key data	Lot, photos, quantity and remaining quantity, price, dates, location, status history
Depends on	Organisations, Catalogue and settings, Grading, File and media storage, Audit log, Notifications
SRS requirements	FR-LOT-01 to FR-LOT-09, BR-01, BR-02

7. Grading and quality

Backend folder: **grading** | Priority: **Must**

Provides the verified quality information that buyers trust. A field agent inspects each lot and records standard measurements and a grade.

Main capabilities	• Grading template for each crop: moisture, foreign matter, damaged share, size or count, overall grade • Agent records measurements, photos and location, and sees the calculated grade before submitting • Works without a connection and syncs later without losing data • Grade calculated from rules that administrators control • Records are permanent; corrections create a new dated record • Administrator approves (lot becomes visible) or rejects with a reason • Delivery-time quality check for comparison with the original grade (Should) • Optional inspection or phytosanitary certificate attachment (Could)
Used by	Field agents (main users), administrators (approval), buyers (delivery check), suppliers (see results)
Key data	Grading record, measurements, grade, photos, location, date
Depends on	Lots, Catalogue and settings, File and media storage, Offline sync, Audit log
SRS requirements	FR-QA-01 to FR-QA-08, NFR-OFF-01

8. Requests

Backend folder: **requests** | Priority: **Must**

Captures what buyers need and lets them browse verified stock.

Main capabilities	• Browse verified lots with filters: crop, grade, quantity, district, price, availability date • Post a request: crop, minimum grade, quantity, target price, delivery location and date • Requests expire after a set period and the buyer is told • Administrator queue of open requests • Reserve a single lot directly without a request (Should)
Used by	Buyers (post and browse), administrators (queue)
Key data	Request, filters, status, expiry
Depends on	Lots, Organisations, Catalogue and settings, Notifications
SRS requirements	FR-REQ-01, FR-REQ-02, FR-REQ-07, FR-REQ-09

9. Offers and matching

Backend folder: **offers** | Priority: **Must**

Turns a buyer request into a single clear offer, including combining stock from several cooperatives (pooling). In V1 administrators do the combining; software assists.

Main capabilities	• Administrator builds an offer from one or more verified lots, including lots from different suppliers • Buyer sees one offer: total quantity, unit price, total price, grades, delivery date and supplier count • Accept, decline or counter, with the full negotiation history stored • Accepting reserves stock from each lot so it cannot be sold twice • Suggested candidate lots ranked by crop, grade, quantity, district and date (Should)
Used by	Administrators (build offers), buyers (respond)
Key data	Offer, offer lines, negotiation history, reservations
Depends on	Requests, Lots, Grading, Orders, Notifications, Audit log
SRS requirements	FR-REQ-03 to FR-REQ-06, FR-REQ-08, FR-ADM-04

10. Orders

Backend folder: **orders** | Priority: **Must**

Manages the agreed purchase from confirmation to settlement. It is the central record that payments, delivery and invoices attach to.

Main capabilities	• Order created from an accepted offer, with a unique order number • Several supplier allocations in one order, shown to the buyer as one order, one delivery and one invoice • Each supplier confirms their allocation; if one declines, the administrator is alerted to substitute stock • Electronic acceptance of terms by buyer and suppliers, stored with who and when • Fee calculation (commission and inspection fee) from administrator settings • Cancellation rules with stock released and a reason recorded • Statuses: Created, Confirmed, Payment Pending, Paid, Dispatched, Delivered, Accepted, Settled, Cancelled, Disputed • Repeat a previous order with one action (Should)
Used by	Buyers, suppliers, administrators
Key data	Order, allocations, price, fees, accepted terms, status history
Depends on	Offers, Lots, Catalogue and settings, Payments, Deliveries, Invoices, Notifications, Audit log
SRS requirements	FR-ORD-01 to FR-ORD-08, section 7.2 of the SRS

Money and fulfilment

11. Payments and settlement

Backend folder: **payments, payouts** | Priority: **Must (highest risk)**

Collects the buyer's money safely before goods move and pays suppliers after delivery is accepted. This module builds trust, so it needs the most careful design and testing.

Main capabilities	• Payment instructions, or payment started through the partner, by mobile money or bank transfer • Payment status received by secure callback or manual confirmation by an administrator, with who confirmed and a reference • An order becomes Paid only after payment is confirmed • Payout calculation: each supplier's share minus Cargo's fee • Payout to the supplier's mobile money number or bank account, with status and reference • Reconciliation against partner statements with differences flagged • Adjustments and partial refunds linked to disputes (Should) • Permanent ledger of every money movement • No card numbers or PINs stored; sensitive steps happen with the partner
Used by	Buyers (pay), suppliers (see payouts), administrators (confirm, reconcile)
Key data	Payment, payout, ledger entry, partner reference, reconciliation status
Depends on	Orders, Integrations (payment partner), Audit log, Notifications, Jobs
SRS requirements	FR-PAY-01 to FR-PAY-09, BR-03, BR-04, NFR-REL-06

12. Deliveries and logistics

Backend folder: **deliveries** | Priority: **Must**

Records how goods move from suppliers to the buyer and what the buyer received. Acceptance here triggers settlement.

Main capabilities	• Administrator records trips: transporter, vehicle, driver contact, pickup and delivery details, planned dates, quantity loaded • Loading confirmation with quantity, time and who confirmed • Buyer confirms receipt: quantity and quality received, with photos for problems • If quantity or quality differs beyond tolerance, the buyer accepts, accepts with adjustment, or opens a dispute • Acceptance triggers settlement • Unanswered deliveries are flagged to an administrator instead of auto-accepted (Should) • Location of delivery confirmation (Could)
Used by	Administrators (trips), buyers (receipt), suppliers (status)
Key data	Trip, loading record, delivery receipt, received quantity and quality, photos
Depends on	Orders, Grading, Payments, File and media storage, Notifications
SRS requirements	FR-DEL-01 to FR-DEL-07

13. Invoicing and records

Backend folder: **invoices** | Priority: **Must**

Produces the paperwork both sides need and keeps a complete record of every trade for audits and tax.

Main capabilities	• Invoice record for each order: buyer, supplier allocations, crop, grade, quantity, unit price, fees, total • Store the EBM invoice or receipt number and a copy of the document, entered by an administrator • Downloadable PDF of order summary, delivery note and invoice details • Full trade record: listing, grading, offer, payment, delivery, acceptance, settlement • Data fields ready so direct EBM integration can be added later
Used by	Buyers and suppliers (download), administrators (enter EBM details)
Key data	Invoice, EBM number, document copy, totals, trade record
Depends on	Orders, Deliveries, Payments, File and media storage
SRS requirements	FR-INV-01 to FR-INV-05

14. Disputes

Backend folder: **disputes** | Priority: **Must**

Gives buyers and suppliers a fair, recorded way to resolve problems such as quality or quantity differences, so trust survives disagreements.

Main capabilities	• Open a dispute with a reason, description and evidence (photos or documents) • Settlement of the affected amount is paused • Administrator reviews both sides, adds notes, decides the outcome (full payment, adjusted payment, refund or other) and closes it • All parties are notified and the decision and reasons are stored • Disputes count in trade histories (Should)
Used by	Buyers and suppliers (open, respond), administrators (decide)
Key data	Dispute, evidence, notes, decision, outcome
Depends on	Orders, Payments, Deliveries, Notifications, Audit log
SRS requirements	FR-DSP-01 to FR-DSP-05

Communication and insight

15. Notifications

Backend folder: **notifications** | Priority: **Must**

Keeps everyone informed at each step so that nothing stalls waiting for someone who did not know.

Main capabilities	• Triggered by key events: account approval, lot verified, new offer, order confirmed, payment received, dispatch, delivery, acceptance, payout, dispute updates, expiring lots and requests • In-app inbox and push notifications, plus SMS for events that need action by users with poor connectivity • Email for buyers and administrators (Should) • Users choose channels for non-critical events (Should) • Message templates in Kinyarwanda and English • Failed messages are retried
Used by	All users
Key data	Notification, event, channel, template, sent and read times
Depends on	Integrations (SMS, email, push), Users, Jobs
SRS requirements	FR-NTF-01 to FR-NTF-05

16. Reports and dashboards

Backend folder: **reports** | Priority: **Must**

Shows how the marketplace is performing, for running the business and for reporting to investors and partners.

Main capabilities	• Administrator dashboard: tons traded, trade value, fee revenue, active suppliers and buyers, open requests, available lots, orders by status • Performance measures: on-time delivery rate, dispute rate, average days from delivery to supplier payout, share of verified stock sold • Supplier and buyer summaries (Should) • Export to CSV or PDF (Should)
Used by	Administrators (full); suppliers and buyers (own summaries)
Key data	Read-only summaries from orders, payments, lots, deliveries and disputes
Depends on	Orders, Payments, Lots, Deliveries, Disputes
SRS requirements	FR-RPT-01 to FR-RPT-04, FR-ADM-07

4. Supporting platform services

These are not about the business itself. They support the business modules, and several of them are shared by all modules.

Service	Folder or location	What it does	Details
Admin console	Admin area of the web app	The single place where administrators run the marketplace. It is a set of screens over the modules above, not a separate backend module.	Users and approvals, lots and grading review, request queue, offers and pooling, orders, payments and reconciliation, deliveries, invoices, disputes, reports, settings and audit log, with search, filters and CSV export
Integrations	Backend: server/integrations	Connects Cargo to outside services through adapters, so a provider can be replaced without changing the modules.	Payment partner (mobile money and bank), SMS, email, push notifications, maps. Verifies signed callbacks and retries failures

Service	Folder or location	What it does	Details
File and media storage	Backend: storage integration	Stores photos and documents securely and links them to records.	Type and size limits, compression before upload, resumable uploads (Should), files never run as code
Background jobs	Backend: server/jobs	Runs tasks on a schedule or in the background.	Expire lots and requests, send and retry notifications, reconcile payments, flag unanswered deliveries
Offline sync	Mobile app	Lets suppliers and especially field agents keep working without a connection.	Local database on the phone, queue of actions, automatic sync with visible status. The server's record of statuses and payments always wins
Localisation	Web, mobile and notification templates	Makes the system usable in Kinyarwanda and English.	Translated screens and messages, RWF currency, kilograms and tons, dates in Central Africa Time

5. How the modules work together

A complete trade passes through the modules in this order:

Step	What happens	Modules involved
1	Supplier and buyer are registered, approved and verified	Authentication, Users, Organisations, Audit log
2	Supplier lists a harvest-ready lot	Lots, Catalogue, File storage
3	Field agent grades the lot, even offline, and an administrator verifies it	Grading, Offline sync, Lots, Audit log
4	Buyer browses stock or posts a request	Requests, Lots
5	Administrator pools lots into one offer and the buyer accepts; stock is reserved	Offers and matching, Lots, Notifications
6	An order is created and everyone accepts the terms	Orders, Catalogue and settings
7	Buyer pays; payment is confirmed before dispatch	Payments, Integrations, Orders
8	Goods are loaded and delivered; buyer records what arrived	Deliveries, Grading, Notifications
9	Buyer accepts; suppliers are paid and invoice records are completed	Orders, Payments, Invoices
10	If there is a problem, a dispute pauses settlement until decided	Disputes, Payments, Audit log
11	The business reviews results	Reports

Notifications, the audit log and background jobs operate throughout the whole flow.

6. Suggested build order

Modules depend on each other, so building in the right order avoids rework. The order below builds the foundation first and the riskiest module (payments) early enough to leave time for partner onboarding and testing.

Stage	Modules	Why this order
A. Foundation	Authentication, Users and roles, Audit log, Catalogue and settings, File and media storage, Integrations base, Localisation	Everything else depends on identity, permissions, reference data and records

Stage	Modules	Why this order
B. Organisations and stock	Organisations and verification, Lots, Grading and quality, Offline sync	Verified stock is the heart of the product, and offline sync must be designed from the start
C. Trading	Requests, Offers and matching, Orders	Turns verified stock into agreed orders
D. Money and delivery	Payments and settlement, Deliveries and logistics, Invoicing and records, Disputes	Needs orders to exist. Start partner onboarding for payments during stage B
E. Communication and insight	Notifications (basic versions are added from stage A), Background jobs, Reports and dashboards, Admin console polish	Completes the experience and the reporting investors expect
F. Release readiness	End-to-end testing, security checks, backup restore test, performance check, store release	Matches the release criteria in the SRS

Within each stage, build the backend part of a module first, then the web or mobile screens that use it, and test it with realistic data before moving on.